

Elemental groups and sets

Within the periodic table, elements are divided into smaller groups and sets that often have similar chemical properties or are related to one another in some other way.

Hydrogen

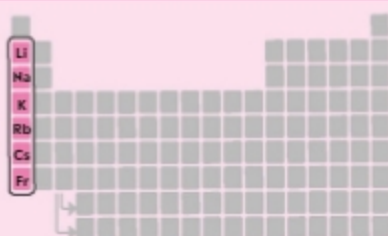


As the most abundant element in the universe, hydrogen is unique. While it is often placed at the top of the alkali metals on the periodic table, hydrogen is a gas at room temperature and has different properties.

Pure hydrogen stored in a glass sphere



Alkali Metals

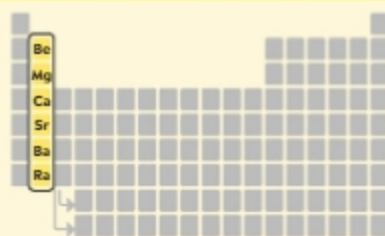


The alkali metals generally have low densities, are soft enough to be cut with a knife, and react violently with both water and air. As a result, they are often stored under a protective layer of oil or in an airless container.

Laboratory sample of pure cesium in an airless vial



Alkaline Earth Metals

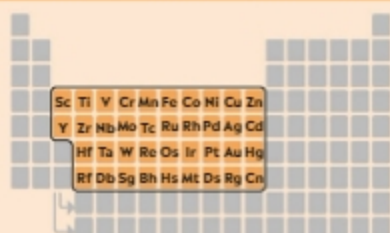


A little less reactive than their group one neighbors, these get their name from the fact that most of them were discovered as oxide compounds in Earth.

Crystals of pure calcium refined in a laboratory



Transition Metals

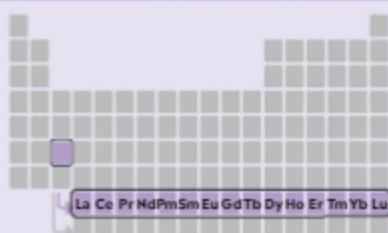


These metals form the largest set of elements in the periodic table. They share many properties with each other, such as their ability to form colored compounds, and are often used in specialized alloys.

Laboratory sample of pure titanium



Lanthanides

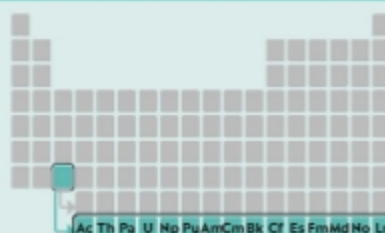


Although they were once known as the "rare earths," the elements in this set are not rare at all. The idea of rarity came from the fact that they are difficult to separate from one another, so they were difficult to find.

Laboratory sample of pure ytterbium



Actinides



Mostly radioactive, actinides are not found commonly in nature, with a few exceptions. Many of them are only produced artificially and have little use beyond research, although uranium is used as fuel in nuclear power plants.

Uranium ore

